Introduction and Problem Motivation



CLIPP

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I. Introduction

As part of the *Climate Integrated Pathways for Podujeva (CLIPP)* project, this competition aims to engage university students in solving realistic decision-making problems related to climate resilience using algorithmic methods. The primary objective of the first competition challenge is to expose participants to a **computationally hard optimization problem** that requires planning, scheduling, and heuristic reasoning under realistic constraints, while remaining solvable within a limited competition timeframe.

To achieve this balance, we build upon the **Street View Routing (SVR)** [1] problem introduced in the inaugural edition of **Google Hash Code**, a team programming competition organized by Google and held annually from 2014 to 2023. Hash Code was explicitly designed to present participants with open-ended, real-world–inspired optimization problems for which exact solutions are infeasible at scale, encouraging the use of greedy methods, heuristics, and metaheuristics rather than closed-form algorithms or exhaustive search.

II. Background: Google Hash Code and Street View Routing

Google Hash Code was created by engineers at Google France as a university-oriented programming competition emphasizing teamwork, practical problem solving, and scalable algorithm design. From its first edition in 2014, Hash Code grew from a local event with a few hundred participants into a global competition with over 100,000 contestants, before being sunset in 2023 as part of Google's broader Coding Competitions program. Throughout its history, Hash Code consistently focused on problems inspired by real engineering challenges faced at Google, particularly those involving large-scale optimization and trade-offs under constraints [2].



Hash Code 2014 in the Google France canteen

The **Street View Routing problem** was the very first problem used in the inaugural Hash Code competition in 2014. It was directly inspired by real optimization work at Google France related to planning routes for Street View cars capturing imagery of urban environments. The original problem models a city as a graph whose vertices represent street junctions and whose edges represent streets, which may be either one-directional or bidirectional. Each street is associated with a traversal time and a length, and all vehicles start from a common depot.

Google Paris
Hash Code

EN

Problem Statement for Main Round (2014-04-05)

Street View Routing





Given a fixed time budget and a fleet of vehicles, the objective is to **maximize the total length of unique streets traversed at least once** by any vehicle. Traversing the same street multiple times, or in both directions, does not yield additional benefit. The problem was provided with multiple datasets of varying scale, ranging from small instances to large city-scale graphs, allowing participants to iteratively improve their solutions during the competition.

The SVR problem strikes a careful balance between **conceptual simplicity and algorithmic depth**. While the input and output formats are straightforward, the combinatorial nature of the problem makes it highly challenging to optimize, especially for large instances. As a result, the problem is particularly well suited for academic competitions, where the emphasis is on reasoning, approximation, and solution quality rather than exact optimality.

III. Computational Complexity and NP-Hardness

From a computational perspective, the Street View Routing problem belongs to a broader family of graph traversal and coverage problems closely related to the **postman problem**. In particular, the version considered here can be viewed as a variant of the **k-mixed postman problem**.

In the classical postman problem, the objective is to find a minimum-cost route that traverses every edge of a graph at least once. Several special cases are known to be tractable: the undirected postman problem and the directed postman problem (both for a single vehicle) admit polynomial-time solutions. However, when the graph contains a mixture of directed and undirected edges, the resulting **mixed postman problem** is known to be **NP-hard**. This hardness result is well established in the literature and has been explicitly discussed in analyses of the Street View Routing problem.

The SVR problem generalizes this setting in multiple ways. First, the city graph is mixed, containing both one-directional and bidirectional streets. Second, instead of a single vehicle, the problem considers **multiple vehicles operating simultaneously**, each subject to its own time constraint. Finally, the objective is not to traverse all edges, but to **select a subset of edges whose total length is maximized under a global time budget**. This selective coverage objective further increases the combinatorial complexity of the problem.

Because the mixed postman problem is NP-hard, and because Street View Routing strictly generalizes it, the SVR problem is also NP-hard. Consequently, computing optimal solutions for realistic instances is infeasible in practice, and effective approaches rely on heuristics, approximation strategies, and metaheuristics. This makes the problem an excellent vehicle for teaching and evaluating algorithmic thinking in the presence of computational intractability [3].

IV. Motivation for Use in the CLIPP Project

We selected the Street View Routing problem as the foundation for this competition for several reasons:

- **Computational difficulty**: The problem is NP-hard, naturally motivating heuristic and approximation-based approaches.
- **Competition-proven design**: It was successfully used as the first problem in a large-scale international programming competition.
- **Clear optimization objective**: The goal is unambiguous and easily measurable, enabling fair and transparent evaluation.
- **Academic suitability**: The problem was originally designed for a university-focused competition and scales well to student-level time constraints.
- **Real-world relevance**: The problem models realistic routing and coverage challenges encountered in urban infrastructure and logistics.

V. Motivation for Adaptation to Climate Resilience

While the original Street View Routing problem focuses purely on maximizing coverage efficiency, it does not explicitly account for environmental constraints or sustainability considerations. In the context of the *Climate Integrated Pathways for Podujeva (CLIPP)* project,

whose objective is to explore algorithmic tools for climate-resilient municipal planning, we extend the original formulation to reflect **resource-aware urban operations under real-world constraints**.



Water scarcity is a growing global challenge and is expected to increasingly affect municipalities worldwide, including the city of Podujeva. As climate change intensifies pressure on water resources, municipal services that depend on water—such as street cleaning and maintenance—must be planned with greater attention to resource efficiency. This motivates the introduction of routing and scheduling problems in which vehicle operations are constrained not only by time and distance, but also by limited and managed resources.

To this end, we adopt the graph-based structure of the Street View Routing problem and introduce several modifications that adapt it to a **realistic street-cleaning scenario**. First, instead of a homogeneous fleet of identical vehicles, the problem features a **heterogeneous fleet** composed of three distinct vehicle types, each with different cleaning capabilities. Second, streets are no longer treated uniformly: they are classified into **mandatory streets** that must be serviced, **optional streets** that yield points when cleaned, and **connector streets** that serve only to enable movement through the network without contributing directly to the score. Third, unlike in the original SVR problem—where vehicles may end their routes at any junction—each vehicle is required to **start from and return to a central depot** within the allotted time, reflecting standard operational constraints of municipal service fleets.

These extensions transform the problem from a purely coverage-based routing task into a **planning and scheduling problem under resource and operational constraints**, while preserving the combinatorial difficulty and heuristic nature of the original formulation. Through this adaptation, participants are challenged not only to design effective heuristics for an NP-hard optimization problem, but also to reason about trade-offs between operational efficiency, service prioritization, and resource conservation—skills that are increasingly central to climate-aware urban planning and decision-making.

References

[1] Google Hash Code Archive. (2014). Problem statement for Main Round Street View Routing (Hash Code 2014 Final Round). GitHub. https://github.com/pierreavn/google-hashcode-archive/blob/main/archive/2014/final/hashcode_2014_final_round.pdf

[2] Pnote.eu. (n.d.). Hashcode. https://pnote.eu/projects/hashcode/

[3] A3nm. (2014, February 28). Google Hashcode 2014. a3nm's blog. https://a3nm.net/blog/google_hashcode_2014.html

[4] DXR. (2022, December 3). Urban-Sweeper S2.0.png [Image]. Wikimedia Commons. https://commons.wikimedia.org/wiki/File:Urban-Sweeper_S2.0.png